

Note: look for notes below

Set up Zabbix Commands :

- sudo -s
- wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_7.0-1+ubuntu24.04_all.deb
- sudo dpkg -i zabbix-release_7.0-1+ubuntu*.deb
- apt update
- apt install zabbix-server-mysql zabbix-frontend-php zabbix-apache-conf zabbix-sql-scripts zabbix-agent

Install MySQL DB :

- sudo apt update
- sudo apt install mysql-server -y
- sudo systemctl status mysql
- sudo mysql_secure_installation
- Set root password (if asked)
- Remove anonymous users (yes)
- Disallow remote root login (yes)
- Remove test database (yes)
- Reload privilege tables (yes)

Create initial database :

- mysql -u root -p (provide root password)
- create database zabbix character set utf8mb4 collate utf8mb4_bin;
- create user user@localhost identified by 'put_your_password';
- grant all privileges on zabbix.* to user@localhost;
- set global log_bin_trust_function_creators = 1;
- quit;
- mysql -u root -p (provide root password)
- set global log_bin_trust_function_creators = 0;
- quit;

Configure the database for Zabbix server :

- vim /etc/zabbix/zabbix_server.conf
- Edit the below :
- DBUser=Provide user name
 - DBPassword=Provide password

DBHost=Provide IP address of DB host

DBName=Provide DB name

- systemctl restart zabbix-server zabbix-agent apache2
- systemctl enable zabbix-server zabbix-agent apache2
- Access the UI to finish the installation http://zabbix_ip_address/zabbix

Install Zabbix agent : (On Target)

- sudo -s
- wget https://repo.zabbix.com/zabbix/7.0/ubuntu/pool/main/z/zabbix-release/zabbix-release_latest_7.0+ubuntu24.04_all.deb
- dpkg -i zabbix-release_latest_7.0+ubuntu24.04_all.deb
- apt update
- apt get install zabbix-agent2

Edit Agent config :

- vim /etc/zabbix/zabbix_agent2.conf
- Edit the below :
- Server=IP Zabbix SRV
 - ServerActive=IP Zabbix SRV
 - HostName=Host Name (Zabbix Server)
- systemctl restart zabbix-agent2
 - systemctl enable zabbix-agent2

Configuration :

Media Type : to send an emails

Alerts > Media types > create media type

Media Type tab :

Name : Email-Gmail

Type : Email

Email Provider : Generic SMTP

SMTP Server : smtp.gmail.com

SMTP Server Port : 587

Email : your email

SMTP helo : hello

Connection Security : STARTTLS

Username : your_username

password : Provide_Token

Message format : HTML

Message template tab:

subject : CPU Alert: {HOST.NAME} - {TRIGGER.NAME}

Message Content/body :

Problem started at: {EVENT.DATE} {EVENT.TIME}

Host: {HOST.NAME}

Trigger: {TRIGGER.NAME}

Severity: {TRIGGER.SEVERITY}

Status: {TRIGGER.STATUS}

Value: {ITEM.VALUE1}

Add Media To receive Emails/Alerts

Users >select user

media tab :

Add >

Send to : put your email

Select severity

CPU Usage :

Add item for cpu

Data Collection > Hosts >select host > items > create item

Item name : CPU Usage

Type : Zabbix Agent (active)

Item key : custom.cpu.usage

Type of information : Numeric (float)

Units : %

Update Intervals : 1m

Config in the target edit “/etc/zabbix/zabbix_agent2.conf” Add custom key

Custom key to monitoring CPU usage

UserParameter=custom.cpu.usage,top -bn1 | grep "Cpu(s)" | awk '{print 100 - \$8}'

Add trigger

Data Collection > Hosts >select host > triggers>create trigger

Name : cpu usage >= 50 %

Select Severity :

Expression : min(/backend/custom.cpu.usage,1m)>=50

Add Action

Alerts > Actions > Trigger Actions > createaction

Name : send email

Conditions : Add > new condition window

type : trigger

Operator : equals

Trigger source : Host

Triggers > select the trigger

Memory usage

Data Collection > Hosts > select host > items > create item

Item name : Memory Usage

Type : Zabbix Agent (active)

Item key : vm.memory.size[pused]

Type of information : Numeric (float)

Units : %

Update Intervals : 1m

Disk usage

Data Collection > Hosts > select host > items > create item

Item name : Disk Space

Type : Zabbix Agent (active)

Item key : vfs.fs.size[/,pused]

Type of information : Numeric (float)

Units : %

Update Intervals : 2m

Disk i/o

Data Collection > Hosts > select host > items > create item

Item name : Disk Reads (ops)

Type : Zabbix Agent (active)

Item key : vfs.dev.read[sda,operations]

Type of information : Numeric (unsigned)

Units : ops

Update Intervals : 1m

Data Collection > Hosts > select host > items > create item

Item name : Disk write (ops)

Type : Zabbix Agent (active)

Item key : vfs.dev.write[sda,operations]

Type of information : Numeric (unsigned)

Units : ops

Update Intervals : 1m

System-Health

Item name : System Health

Type : zabbix agent (active)

Key : system.health

Type of information: text

Update interval: 5m

Config in the target edit “/etc/zabbix/zabbix_agent2.conf” Add custom key

Custom key to monitoring System Health

UserParameter=system.health,systemctl is-system-running 2>/dev/null || echo unknown

Dashboard :

Dashboards > create dashboard

Select Owner :

Name : Linux Project

Add a Page for each VM

Database Page:

Add → Type → Gauge

Main Configuration

Name: CPU Usage Per %

Refresh Interval: 30Seconds

Item:db:CPU Usage

Min: 0

Max 100

Repeat for Disk – Memory

Add → Type → Graph

Name: Disk i/o

Refresh Interval: 1minute

Data Set main Configuration :

select host: db

item patterns: Diske Reads (ops) Disk Write(ops)

Add → Type → Host availability

Name: Host availability

Refresh interval: 10 minutes

Host Groups : ubuntu

Interface type: zabbix agent (active checks) & zabbix agent (passive checks)

Add → Type → Data overview

Name : Data

Host Group: ubuntu

Hosts : db

Host Location: Top

Notes :

- 1- Repeat Trigger & Action Steps in critical items that you need to receive a notification for
- 2- Zabbix SRV OS & Target OS = Ubuntu
- 3- Web Server = Apache
- 4- Repeat step "Install Zabbix agent" for each target. Instead, you can use an Ansible playbook
- 5- You have a choice to select a predefined dashboard

LinkedIn: <https://www.linkedin.com/in/ahmed-mahmoud-abuzaid-49a800226/>